TEAM VAJRA PRESENTS

THE KIRANA PROBLEM

Stock and reorder for a shop that has never used software.

बिना सॉफ़्टवेयर वाली दुकान का स्टॉक और रीऑर्डर

GDGoC SRMCEM × MLH

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Illustrative app preview

PROBLEM & SOLUTION

NOT A BILLING SYSTEM

THE PROBLEM

- Stock lives in a notebook or the owner's head
- Dead stock locks up cash for months
- Fast movers run out and customers walk away
- Billing apps need a PC, training and a fee

DEAD STOCK ⇄ STOCK-OUTS

OUR SOLUTION

An offline, Hindi-first phone app that tells the owner exactly what to reorder.

- **Add stock and log a sale in a few taps**
- **Low-stock and not-moving lists: a plain reorder sheet**
- **Works offline; syncs to Firebase when online**
- **Camera, barcode and expiry scan, plus voice entry**
- **Self-learning reorder and festive demand forecast**
- **WhatsApp order list to the distributor**

TECHNICAL APPROACH

FRONTEND

TypeScript · React · Tailwind CSS

CLOUD **GOOGLE**

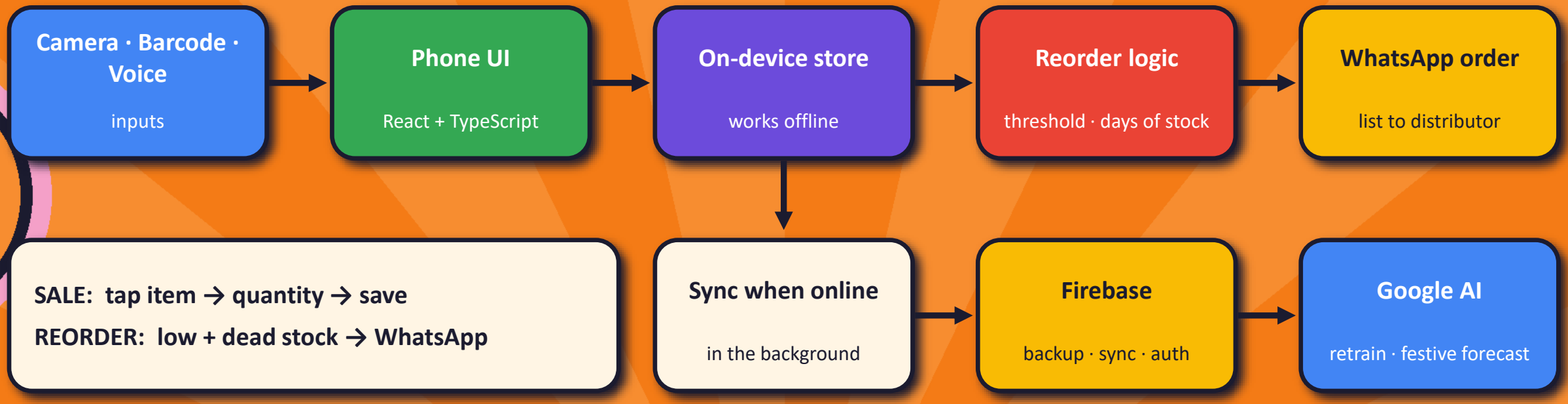
Firebase backup and sync

AI **GOOGLE**

Google AI APIs for ML reorder

OFFLINE

On-device storage, works with no signal



The updated ML model returns to the phone on the next sync

FEASIBILITY & VIABILITY

₹0
monthly fee

0
hours of training

0
bars of signal needed

1
hand needed to use it

RISK	MITIGATION / BACKUP
No internet	Offline-first: data stays on the device, syncs later
Dozens, half kilos, loose pieces	Decimal quantities, unit-aware
Low literacy and language	Hindi + English, big icons, voice entry
Little data for the ML model	Falls back to fixed thresholds
Phone lost or cleared	Firebase backup restores the shop

BUILT TO RUN ON A BASIC PHONE

IMPACT & BENEFITS

DESIGN TARGET: FASTER THAN A NOTEBOOK

FOR OWNERS

- Cash freed from dead stock
- Fewer missed Saturday-evening sales
- Expiry alerts cut chemist losses

FOR DISTRIBUTORS

- Clear order lists on WhatsApp
- Fewer calls and mix-ups

FOR SOCIETY

- Small retail goes digital without a PC
- Hindi-first digital inclusion
- Low cost, no subscription

SCALES BEYOND THE HACKATHON

More languages | More shop types | Distributor integration | Low cloud cost

DEMO & NEXT STEPS

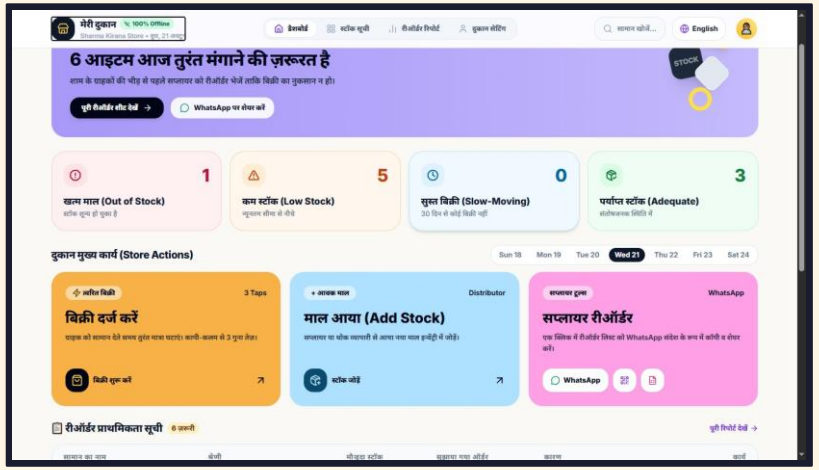
3-STEP LIVE DEMO

1. Log a sale in aeroplane mode
2. Open the reorder list
3. Send the WhatsApp order

REFERENCES

- Firebase: offline data, sync
- Google AI: Gemini API docs
- WhatsApp: Click to Chat
- MDN: PWA, service workers

LIVE DEMO



FUTURE SCOPE: more Indian languages · deeper distributor WhatsApp integration · more shop types

THANK YOU · TEAM VAJRA · Aditya Nath Patel · Saumy Chaurasia · Samarth Singh